A273 – The Reckoning Bead

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Abstract. Computing with tokens — abacus beads, shells, coins — is the limiting case in which Landauer's argument comes apart into its two halves. The *accounting* holds there exactly and is for once visible: the stock from which the tokens come and into which they flow back is a bath one can point at. The *thermal conversion* $\Delta S \to Q = T\Delta S$ does not hold there: an abacus bead costs about 5×10^{15} times the Landauer bound to move, its positional bit carries 4×10^{-23} of its own thermal entropy, and its positional barrier stands at $3.6 \times 10^{15}\,kT$. The token computer therefore falls squarely under the very restriction Landauer himself stated and the reception dropped (A272): the counted states are not a thermal ensemble. From this follows an ordering that runs against intuition: **visibility of the structure and binding force of the number run in opposite directions.** The cruder the carrier, the clearer the accounting and the more irrelevant the bound. The passage back is continuous — shrink the token until its barrier approaches kT and it becomes a Brownian reckoning bead, which is Bérut's colloid.

Layer markings. **[SETZUNG]** declared starting point (German corpus token: [SETZUNG]) — **[B]** derived algebraically/mathematically from declared foundations — **[Q]** source — external primary source or measured value — **[S]** plausibility sketch — **[H]** open research question.

Check script. `a273_rechenkugel.py` (checks 1–14). Every number in this document is backed there by assertions; the parameter choices are marked in the script as stipulations and can be varied freely.

.1 The stock as a visible bath

A271 works with the world ocean: a glass is poured in, the volume does not vanish, but nobody gets it back. The image is correct and has one drawback — the bath has to be inferred. Nobody sees the 10^{23} degrees of freedom into which the bit volume tips.

[SETZUNG] With token computation this is different. An abacus, a heap of shells, a till with coins: next to the working area there lies a **stock**. The tokens come from it and flow back into it. The stock is a bath one can point at.

```
 Working area              Stock (bath)
+-----------------+       +-------------------------+
| o o . o . . o . |       | o o o o o o o o o o o o |
| . o o . o o . o |  ---> o o o o o o o o o o o o |
+-----------------+       +-------------------------+
 W = 2^N config.           tokens conserved, arrangement mixed
```

[B] The structure is step for step the same as in A271:

- **Region.** A token in the left position and the same token in the right position are two distinguishable macro-regions.
- **Region reduction.** Clearing the board maps all configurations onto one — all tokens in the stock. Non-bijective, exactly like $\{A, B\} \to Z$.
- **Conservation.** No token vanishes. They spread into an arrangement from which the previous configuration can no longer be read off. That is the Liouville analogue — and here it is literally countable.
- **Content neutrality.** The stock does not know whether a token stood for a debit, for a five, or for nothing. It absorbs it the same way.

[B] Cash with coins and a till is the same figure, and the double meaning of the word "accounting" stops being a metaphor here: the balance of a till and phase-space accounting have the same form. Barter is the case without tokens — there is no stock there and correspondingly no region reduction, only bijective exchange.

.2 The accounting holds exactly

[B] For N tokens with two positions each, before clearing $W_{\text{before}} = 2^N$ and after $W_{\text{after}} = 1$, hence

$$\frac{\Delta S}{k_B} = \ln \frac{W_{\text{before}}}{W_{\text{after}}} = N \ln 2. \tag{1}$$

This is exact and carrier-independent *(check script, check 10)*. It holds for beads, for shells, for transistors, for colloids. Nothing in equation (1) demands that the token be small, light, or thermally driven. The count is combinatorics over regions — exactly what A271 identifies as primary.

[B] Up to this point the token computer is a complete model of Landauer accounting, and a better one than any other, because every step can be seen.

.3 The bound does not hold

[SETZUNG] For the following numbers it is stipulated: an abacus bead of mass $m = 0.5$ g, a displacement $d = 1$ cm, a friction coefficient $\mu = 0.3$, an ambient temperature $T = 300$ K, a molar mass of 100 g/mol and a molar entropy of 50 J/(mol·K) for a solid. All stipulations are marked in the check script and can be changed; the orders of magnitude do not depend on their precise values.

[B] The work per bead displacement. Sliding friction gives $W = \mu m g d = 1.47 \times 10^{-5}$ J *(check script, check 2)*. The Landauer bound at 300 K is $k_B T \ln 2 = 2.87 \times 10^{-21}$ J *(check script, check 1)*. The ratio is

$$\frac{W_{\text{bead}}}{k_B T \ln 2} \approx 5.1 \times 10^{15} \qquad \textit{(check script, check 3)}. \tag{2}$$

The scale band

Carrier	Cost per switching event	Multiple of $k_B T \ln 2$
colloid, quasistatic [B4][B5]	$\sim 3 \times 10^{-21}$ J	1–10
CMOS transistor (A271)	10^{-17}–10^{-16} J	$3.5 \times 10^3 - 3.5 \times 10^4$
abacus bead	1.5×10^{-5} J	5.1×10^{15}

[B] Between bead and CMOS lie roughly **eleven decades** *(check script, check 5)*. The ordering is thus the reverse of the obvious conjecture: Landauer binds not at the primitive end but at the small one. The cruder the carrier, the further the bound lies below the actual expenditure.

Why the bead pays nothing

Two numbers account for the finding, and both concern the bead itself, not the accounting.

[B] First: the positional bit is vanishing against the bead's own entropy. A bead of 0.5 g carries, under the stipulations above, an entropy of 0.25 J/K, that is $1.8 \times 10^{22}\, k_B$ *(check script, check 6)*. The positional bit carries $\ln 2 = 0.69\, k_B$. Its share is

$$\frac{\ln 2}{S_{\text{bead}}/k_B} \approx 3.8 \times 10^{-23} \qquad \textit{(check script, check 7)}. \tag{3}$$

Moving the bead changes essentially nothing in the thermally accessible phase space. The region reduction takes place in a coordinate decoupled from the bead's 10^{22} thermal degrees of freedom.

[B] Second: the positional barrier is thermally insurmountable. It stands at $3.6 \times 10^{15}\, k_B T$ *(check script, check 8)*. An Arrhenius lifetime $\propto \exp(3.6 \times 10^{15})$ is not numerically representable. There is no thermal transfer between "left" and "right", and therefore no equilibrium ensemble over those two states.

[B] The bead is thus the extreme case of the row "emergent two-valuedness" in A271: the two is a partition imposed on a quasi-continuum of 10^{22} microstates, and it survives not because it is fundamental but because the barrier is absurdly high.

.4 The two halves of Landauer's argument

[B] The token computer thereby does something no other model does as cleanly: it separates the two steps of which Landauer's argument consists, and shows that they are independent.

Step	Statement	Presupposition
combinatorics	$\Delta S = k_B \ln(W_{\text{b}}/W_{\text{a}})$	regions are distinguishable and countable
conversion	$Q = T\Delta S$	the counted states are thermally occupied

[B] The first step holds exactly for the bead. The second does not hold for it at all. The usual presentation runs the two together in one move and thereby makes the second presupposition invisible. Only when a carrier violates it maximally does it come into view.

> **[B]** Landauer's bound is not the statement "region reduction costs". It is the statement "region reduction *in thermally occupied states* costs". The qualifier is not cosmetic — it carries the whole conversion.

[B] This sharpens A271 at a place where it needs to be more precise. A271 §6 requires, for classical systems, distinguishability *above* the thermal resolution limit. That is the condition for two regions to count as two. It is *not* the condition for their reduction to cost heat — for that one additionally needs the barrier *not* to lie arbitrarily far above $k_B T$. The two conditions point in opposite directions, and between them lies the narrow band in which Landauer binds.

.5 Landauer's own restriction

[Q] The finding is not a new discovery but the extreme case of a restriction Landauer stated himself in 1961: his reset operation was applied to a thermal equilibrium ensemble; for information not yet thermalised the calculation does not apply directly [B1]. A272 documents the passage and shows that the reception dropped it.

[B] With the abacus the restriction is maximally violated, and in two ways at once:

1. **No thermal ensemble.** The distribution over "left" and "right" does not arise from thermal occupation but from the hand that placed the bead.
2. **Complete knowledge.** Whoever placed the beads knows the configuration. In the feedback bound (A272) this makes $I = \ln 2$ per token and

$$Q_{\text{min}} = k_B T(\ln 2 - I) = 0 \qquad \textit{(check script, check 11)}. \tag{4}$$

[B] Thermodynamically, then, clearing a known abacus costs nothing. What it does cost is friction — and friction is an engineering quantity with no lower bound in nature, reducible at will by better bearings.

.6 The passage back: the Brownian reckoning bead

[B] The passage back into Landauer's domain of validity is continuous and can be computed. Setting the positional barrier equal to $k_B T$ gives, for the critical token mass,

$$\mu m g d = k_B T \quad \Longrightarrow \quad m_{\text{crit}} = \frac{k_B T}{\mu g d} \approx 1.4 \times 10^{-19}\,\text{kg} \qquad \textit{(check script, check 9)}. \tag{5}$$

[B] For comparison: a silica colloid of 2 µm diameter weighs 8.4×10^{-15} kg, roughly 6×10^4 times more *(check script, check 9)*. But it holds its position not by friction, rather by an optical trap, and the height of that barrier is adjustable down to a few $k_B T$. That is exactly what makes it a Brownian reckoning bead — and exactly why Bérut's apparatus [B4] is the only one in which the bound actually binds.

> **[B]** Bérut's colloid is not a model of the bit. It is an abacus bead small enough for the bath to push it.

[B] The series thereby closes: shell, coin, abacus bead, transistor, colloid. It is a scale, not a jump. What changes along the scale is not the accounting — it is the same everywhere — but the ratio of the positional barrier to $k_B T$.

.7 Two axes, running opposite

[B] From this follows the ordering that gives the document its yield:

Carrier	Visibility of the structure	Binding force of the bound
abacus, shells, coins	complete (bath can be pointed at)	none (10^{15} above)
CMOS	slight (bath is the lattice)	weak (10^4 above)
colloid in an optical trap	none (bath not separable)	full (factor 1–10)

> **[B]** At the most primitive level the accounting is maximally visible and the bound maximally irrelevant. Visibility of the structure and binding force of the number are two different axes, and they run in opposite directions.

[B] This also explains why intuition regularly misleads in this matter. Anyone wanting to make Landauer plausible reaches for the coarse image — the bead, the drawer, the glass of water — and thereby lands exactly where the bound has ceased to mean anything. The image carries the structure and loses the number.

.8 Two floors: $\hbar c/L$ and k_BT

[B] The thermal floor k_BT is not the only one. There is a second, quantum-geometric floor: $\hbar c/L$, the energy of a mode of extension L. Both limit how finely a state can be pinned down at all — but they are of different origin, and the larger of the two governs.

[B] The quantum-geometric floor $\hbar c/L$ is the energy of a mode of extension L. It applies to massless or field-like carriers — photon, winding mode — and is temperature-independent. It says how finely a state can be pinned down at all at a given extension.

[B] The thermal floor k_BT applies wherever a carrier couples to a bath. It says how deep a well must be to hold against fluctuations — and it is what decides about Landauer.

[B] The larger of the two governs. This sorts the cases:

- **Massive mechanical carriers.** For bead and colloid, $\hbar c/L$ is not the relevant quantum scale at all; the relevant one would be $\hbar^2/(2mL^2)$, and that amounts to $3 \times 10^{-41}\, k_BT$ for the abacus bead and $2 \times 10^{-22}\, k_BT$ for Bérut's colloid *(check script, check 12)*. The quantum-geometric floor lies so far beneath everything here that only the thermal floor counts — the criterion of this document is the whole story there.
- **Field-like carriers.** There $\hbar c/L$ carries. At 300 K the two floors cross at $L = \hbar c/(k_BT) = 7.6$ µm *(check script, check 13)*; below that length the quantum-geometric floor dominates, above it the thermal one.
- **The qubit** is the case in which the quantum-geometric floor exceeds the thermal one: $hf/k_BT = 24$ at 5 GHz and 10 mK *(check script, check 14)*. That is exactly why it behaves quantum-mechanically at all — and exactly why it lies close enough to the thermal band to need error correction.

[B] Landauer therefore binds where the thermal floor leads *and* the barrier lies within a few k_BT. Where the quantum-geometric floor leads, $k_BT \ln 2$ is not the relevant quantity. In that sense, but only in that sense, the bound is a special case: it holds for a class of carriers, not for all of them.

.9 The token stock as a cost category of its own

[B] One point remains that thermodynamic accounting does not capture and that becomes especially clear with the token computer: **without tokens one cannot compute.** The stock is an exhaustible resource. An abacus with too few beads does not compute more slowly; it does not compute at all.

[B] This is a *material bound*, not a thermodynamic one. It has no lower limit in k_BT; it has a lower limit in the number of items. It is related to what A271 §7.3 opens up — the computer pays for existing, not only for computing — but it is not the same thing: there the subject is a running entropy current, here a standing inventory.

[S] With semiconductors this category drops out of view, because the stock is cast in silicon: the regions are laid down in fabrication and not drawn during operation. The inventory is there, so it is not counted. On the abacus it is visible, because it can be used up.

[H] Open question. Whether inventory and current can be carried in a single ledger — token stock and entropy current as two balances of one computer — is open. The question is the same as the one at the end of A272, posed from the material side.

.10 Placement in the corpus

[B] The three pieces stand to one another as follows:

- **A271** does the accounting: region, region reduction, Liouville, bath, bound. The physical core.
- **A272** does the textual criticism: what Landauer proved in 1961, his own ensemble restriction, the reception history, the language.
- **A273** exhibits the limiting case at which both findings converge and become visible: the token computer satisfies A271's accounting exactly and violates A272's ensemble condition maximally.

[B] The dictionary of A272 §1.1 applies unchanged. In the language of this document: token = carrier = region; stock = bath; clearing the board = carrier operation = region reduction.

[B] Contact with FFGFT: none. Like A271 and A272, this document lives at the level of statistical mechanics; the ξ scheme is untouched.

.11 Layer-status summary

Statement	**Status**
The token stock is a bath analogue one can point at	**[SETZUNG]**
Parameters of the abacus bead (mass, distance, friction, T)	**[SETZUNG]**
Clearing the board is a non-bijective region map	**[B]**
Tokens are conserved (Liouville analogue)	**[B]**
$\Delta S/k_B = N \ln 2$, exact and carrier-independent	**[B]**
$k_BT \ln 2 = 2.87 \times 10^{-21}$ J at 300 K	**[B]**
Bead displacement 1.47×10^{-5} J	**[B]**
Bead $\approx 5.1 \times 10^{15} \times$ Landauer	**[B]**
Scale band colloid / CMOS / bead	**[Q]**
Eleven decades between bead and CMOS	**[B]**
Bead's own entropy $1.8 \times 10^{22}\, k_B$	**[B]**
Positional bit = 3.8×10^{-23} of that entropy	**[B]**
Positional barrier $3.6 \times 10^{15}\, k_BT$	**[B]**
Combinatorics and thermal conversion are independent	**[B]**
The conversion presupposes thermally occupied states	**[B]**
Landauer's ensemble restriction (original passage)	**[Q]**

Statement	Status
Known abacus: $I = \ln 2$, hence $Q_{\min} = 0$	**[B]**
Critical token mass 1.4×10^{-19} kg	**[B]**
Silica colloid 2 µm: 8.4×10^{-15} kg	**[B]**
Bérut's colloid as a Brownian reckoning bead	**[B]**
Visibility and binding force run in opposite directions	**[B]**
Token stock as a material bound	**[B]**
Why the category drops out of view with semiconductors	**[S]**
A single ledger for inventory and current	**[H]**

Balance: 17× **[B]** · 2× **[Q]** · 2× **[SETZUNG]** · 1× **[S]** · 1× **[H]** = 23 entries.

.12 Use of AI tools

This document was produced with the aid of AI-assisted tools. The questions, the stipulations and all decisions of content are the author's; the tools draft the prose, check the arithmetic and produce the verification scripts. Responsibility for content and for errors rests entirely with the author. The full wording is given in A010.

.13 Check script

- `python/A_Serie_Skripte/a273_rechenkugel.py` (checks 1–14)

.14 What remains open

Whether inventory and current can be carried in a single ledger — token stock and entropy current as two balances of one computer — is open.

.15 Supersedes

This document has no direct counterpart in the older corpus; it works out the limiting case for A271 and A272.

Sources

[B1] R. Landauer, *Irreversibility and Heat Generation in the Computing Process*, IBM J. Res. Dev. **5**, 183 (1961).

[B2] Liouville's theorem: standard result of Hamiltonian mechanics; e.g. Landau/Lifshitz, *Mechanics*, §46.

[B3] C. H. Bennett, *The Thermodynamics of Computation — a Review*, Int. J. Theor. Phys. **21**, 905 (1982).

[B4] A. Bérut et al., *Experimental verification of Landauer's principle*, Nature **483**, 187 (2012).

[B5] Y. Jun, M. Gavrilov, J. Bechhoefer, *High-Precision Test of Landauer's Principle in a Feedback Trap*, Phys. Rev. Lett. **113**, 190601 (2014).

[B6] T. Sagawa and M. Ueda, *Nonequilibrium Thermodynamics of Feedback Control*, Phys. Rev. E **85**, 021104 (2012).

[B7] Semiclassical state counting Γ/h^f and the entropy of solids: standard; e.g. Landau/Lifshitz, *Statistical Physics*, §7 and §64.

[B8] CODATA 2018 values; $k_B = 1.380649 \times 10^{-23}$ J/K exactly defined since the SI revision of 2019.

The numerical values in this document come from the check script `a273_rechenkugel.py`; the parameters set there are model choices, not measured values. The orders of magnitude are insensitive to their variation.